3.2	**The sources, origins, physical and working properties of papers and boards and their social and ecological footprint**	To apply knowledge and understanding of the advantages, disadvantages and applications of the following materials, in order to be able to discriminate between them and select appropriately.
		3.2.1 Paper: a copier paper (in topic 1) b cartridge paper (in topic 1) c tracing paper (in topic 1) d bond e heat transfer printing paper (sublimation printing).
		3.2.2 Board: a folding boxboard (in topic 1) b corrugated board (in topic 1) c solid white board (in topic 1) d foil-lined board.
		3.2.3 Packaging laminate (including Tetra Pak™): a paperboard b polyethylene c aluminium foil.
		3.2.4 Sources and origins – where paper and boards are resourced/manufactured and their geographical origin: a China, USA, Japan – pulp, paper and cardboard b Eastern Asia – rice paper.
		3.2.5 The physical characteristics of each type of paper and board: a density b transparency c texture.
		3.2.6 Working properties – the way in which each material behaves or responds to external sources: a flexibility (in topic 1) b printability (in topic 1) c biodegradability (in topic 1) d weight e surface finish f printability g absorbency.

PAPERS AND BOARDS

Key idea	What students need to learn:
	3.2.7 Social footprint: a trend forecasting b impact of material production on communities and wildlife c impact of logging and material production on communities and wildlife d recycling/disposal – ethical responsibility e reduction of packaging materials – reduction in litter/waste/energy use f brand identity – consumerism, changing the packaging of products over time.
	3.2.8 Ecological footprint: a sustainability b harvesting and erosion of the landscape c processing d transportation e wastage f pollution.